Type MS · Explain water movement by osmosis in body contexts

Answer 1 of 6 · Question 8 part c · 3 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

8 (a)	• provides nutrients/named nutrient/oxygen to body cells; • removes waste/named waste from body cells;		2
(b)	• capillaries have small gaps/pores; • water/small molecules/named small molecule forced out; • due to high (hydrostatic) pressure (in capillaries); • at arteriole end;	ignore plasma allow due to high pressure (in capillaries) allow artery	4
(c)	• osmosis; • lower water potential inside cells/higher water potential in tissue fluid;; • water moves from high to low water potential/down a water potential gradient;		3
(d)	• swelling/increase in cell volume; • cells burst/lyse;		2

Total Marks 11

Type MS · Explain water movement by osmosis in body contexts

Answer 1 of 6 · Question 8 part c · 3 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

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Type MS · Explain water movement by osmosis in body contexts

Answer 2 of 6 · Question 5 part c.iii · 4 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c.iii in this mark scheme extract

5(a)	movement of a substance against a concentration gradient/low to high concentration; using energy/ATP;		**(2)**

Question Number	Answer	Additional guidance	Mark
5(b)	• cut pieces of potato to same size/length/mass; • weigh each piece; • place into different solutions; • leave for at least 30 minutes; • remove and dry; • re-weigh/re-measure length; • determine which concentration has no / least change in mass/length;		**(5)**

Question Number	Answer	Additional guidance	Mark
5(c)(i)	0.10 × 100; 7.90 1.3;	Ecf allow correct answer to incorrect figures used in calculation Allow 1.26 / 1.27 for final correct answer	**(2)**

Type MS · Explain water movement by osmosis in body contexts

Answer 2 of 6 · Question 5 part c.iii · 4 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c.iii in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
5(c)(ii)	initial mass/volume/size varies; greater change in mass/volume if potato is bigger; provides a standardised result; allows comparison;		**(2)**
Question Number	Answer	Additional guidance	Mark
5(c)(iii)	• plot a graph; • concentration vs percentage change; • find (from graph) concentration at which percentage change is zero/lowest change; • this is the concentration of the contents;		**(4)**

Total 15 Marks

Type MS · Explain water movement by osmosis in body contexts

Answer 2 of 6 · Question 5 part c.iii · 4 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c.iii in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain water movement by osmosis in body contexts

Answer 3 of 6 · Question 7 part b.ii · 2 marks

2020 JAN · Paper 01 · find part b.ii in this mark scheme extract

Question number					
7	(a)	(i)	• 67.5 ÷ 48.3 (1) • 1:1.4 (1)	Full marks for correct final answer	2
		(ii)	Any two from • more males than females are overweight in all countries (1) • lowest percentage of people between the ages of 18-24 in all countries are overweight (1) • increased in all countries 18 - 64 (1)		2
		(iii)	sample size too small (1) not all overweight people identified/sample not representative (1)		2
	(b)	(i)	Any three from: • pancreas (1) • releases insulin (1) • reference to the liver taking up glucose (1) • glucose converted to glycogen (1)	R sugar R breaks down glucose	3
		(ii)	Any two from: • high blood glucose increases the concentration of the blood (1) • water moves out of cells (1) • by osmosis (1) • excess water in blood excreted in urine (1)		2

Total for Question 7 = 11 marks

Type MS · Explain water movement by osmosis in body contexts

Answer 4 of 6 · Question 6 part d · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

6	(a)	(i)	A; (brain)		1
		(ii)	B; (endocrine)		1
	(b)	(i)	• 75 (from weight of man) (1) • 320 × 2 (extra from drinking beer) (1) • 715cm^3 (1)		3
		(ii)	• 715 - 75 (1) • (640 ÷ 75) × 100 (1) • = 853% (1)		3
	(c)		any two from • increases permeability of collecting duct (1) • more water reabsorbed (1) • into blood (1)		2
	(d)		any two from • red blood cells loses water (1) • become distorted (1) • less oxygen transported (1) • volume of plasma decreases (1) • blood becomes thicker/more difficult to circulate(1) • increase in blood pressure (1)	Ignore less water in blood Allow water potential decreases	2
	(e)		any three from • waste products continue to be produced (1) • toxic (1) • need to be removed in urine (1) • less reabsorption of water (1) • less ADH produced (1)	Allow named waste product(s)	3

Total 15 marks

Type MS · Explain water movement by osmosis in body contexts

Answer 4 of 6 · Question 6 part d · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks
		Allow reverse for whole	

Type MS · Explain water movement by osmosis in body contexts

Answer 5 of 6 · Question 5 part e · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part e in this mark scheme extract

5 (a)	• two monosaccharides; • joined together;		2
(b)	molecule too large;		1
(c) (i)	• transcription (of lactase gene) does not occur; • RNA is not produced/synthesised; • lactase/protein/enzyme not produced;		3
(ii)	• lactose can't be digested; • into components to be absorbed;		2
(d)	• they eat more dairy produce; • being able to digest it would give them an advantage;		2
(e)	• water passes from cells; • into gut; • diarrhoea causes poor absorption of vitamin D and Ca^{2+}; • as they are water soluble;		4

Total 14 marks

Type MS · Explain water movement by osmosis in body contexts

Answer 5 of 6 · Question 5 part e · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part e in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain water movement by osmosis in body contexts

Answer 6 of 6 · Question 8 part d · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

8 (a)	• provides nutrients/named nutrient/oxygen to body cells; • removes waste/named waste from body cells;		2
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Total Marks 11

Type MS · Explain water movement by osmosis in body contexts

Answer 6 of 6 · Question 8 part d · 2 marks

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